ASSIGNMENT COVERSHEET

UTS

UTS: ENGINEERING & INFORMATION TECHNOLOGY

SUBJECT NUMBER & NAME	NAME OF STUDENT(s) (PRINT CLEARLY)	STUDENT ID(s)
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NAME OF TUTOR	TUTORIAL GROUP	DUE DATE
Sylvan Rudduck	4	27 August

ASSESSMENT ITEM NUMBER & TITLE

Assessment Task 1: Data Visualisation Foundations

☐ I acknowledge that if AI or another nonrecoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used.
☐ I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.
☐ I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.
☐ I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

Declaration of originality: The work contained in this assignment, other than that specifically attributed to another source, is that of the author(s) and has not been previously submitted for assessment. I have rewritten any material provided by AI or other nonrecoverable sources and where appropriate acknowledged their contribution. I understand that, should this declaration be found to be false, disciplinary action could be taken and penalties imposed in accordance with University policy and rules. In the statement below, I have indicated the extent to which I have collaborated with others, whom I have named.

No content generated by AI technologies or other sources has been presented as my own work and I have rewritten any text provided by AI or other sources in my own words.

Statement of collaboration:

Signature of student(s) ____Warit Srichairattanakul_____ Date ___27 August 2025_____

ASSIGNMENT RECEIPT

To be completed by the student if a receipt is required

SUBJECT NUMBER & NAME	NAME OF TUTOR

SIGNATURE OF TUTOR	RECEIVED DATE

STYLE GUIDE for ASSIGNMENT SUBMISSION

Before submitting an assignment, you should refer to the policies and guidelines set out in the following:

- FEIT Student Guide
- UTS Library - referencing
- HELPS - English and academic literacy support
- UTS GSU - coursework assessment policy and procedures

Unless your Subject Coordinator has indicated otherwise in the Subject Outline, you must follow the instructions below for submission of assignments in the Faculty of Engineering and Information Technology.

Writing style

It is usually best to write your initial draft in the default settings of your software without formatting. Use the following guides in your writing.

Purpose and audience: use the correct genre and language style expected for the particular task.

Language: use 'plain English' for all technical writing. More information about this language style can be found at www.plainenglish.co.uk/free-guides.html.

Use spelling and grammar software tools to check your writing. Edit your document.

Standards: always use:

- Australian spelling standards (Macquarie Dictionary)
- SI (International System of Units) units of measurement
- ISO (International Organisation for Standardisation) for writing dates and times for international documents. For example **yyyy-mm-dd** or **hh-mm-ss**. However, for most applications it is more helpful to present the date in full as **26 August 2016**.

Graphics and tables should:

- be numbered
- have an appropriate heading and/or caption
- be fully labelled
- be correctly referenced.

Presentation

Unless otherwise instructed, all assignment submissions should be **word processed** using spell-check and grammar-check software. Work should be well **edited** before submission. Use the following default settings:

Page setup: set margins at no less than 20mm all around.

Paper: print on A4 bond, double-spaced and preferably double-sided, left justified.

Font: use the software default style to provide consistency. The recommended style includes:

- 10-12 pt font
- consistent formatting with a limited number of fonts
- lines no more than 60 characters (use wider margins or columns if you need to make lines shorter)

Header should include:

- your name and student number
- the title of the paper or task.

Footer should include the page number and current date.

Cover sheet and statement of originality: all work submitted for assessment must be the original work of the student(s) submitting the work. A standard faculty cover sheet (see over) must be attached to the front of the submission. Any collaboration between the submitting student and others must be declared on the cover sheet.

Referencing

All sources of information used in the preparation of your submission must be acknowledged using the APA system of referencing. This includes all print, video, electronic sources.

Phrases, sentences or paragraphs taken verbatim from a source must be in quotation marks and the source(s) cited using both **in-text** referencing and a **reference list**.

Plagiarism is the failure to acknowledge sources of information. You should be fully aware of the meaning of plagiarism and its consequences both to your marks, position at the university and criminal liability. The plagiarism in your assignment submissions can be assessed both in hard copy and in soft copy through software such as Turnitin.

The UTS Library and UTS HELPS (web links above) provide extensive information for students on referencing correctly to support you in avoiding plagiarism.

Data collection and preparation process

The dataset for this analysis was collected from the company's historical market data records published by the Australian Securities Exchange (ASX), covering the period from 1 July 2024 to 30 June 2025. The dataset was obtained in Excel format, which initially contained string values, general numeric values, and unformatted columns. The raw data required transformation into a structured format before it could be used for analysis.

The dataset includes key financial attributes such as Date, Open Price, High Price, Low Price, Close Price, Trading Volume, Market Capitalisation, and Shares on Issue. These attributes were chosen because they provide the necessary foundation for stock performance and valuation analysis.

To prepare for visualisation and analysis, I convert the Date from string format to a proper date format for time-series analysis. Then check for any missing values in the dataset to ensure data consistency and completeness. Finally, I sorted the dataset in ascending order of Date to maintain a logical sequence for trend analysis. Then, I convert the format of Open, High, Low, Close from a general numeric format to a currency format to ensure that financial values are displayed appropriately.

Additionally, I added a dividend column and set it to 0 to indicate that SGR did not pay dividends to investors during this period. Price to Earnings (SGR) column for reflecting the company's Price-to-Earnings ratio, and Price to Earnings (ASX Average) column fixed at 18.0, used as a benchmark for comparison.

After reviewing the dataset, I found that from 30 August 2024 to 26 September 2024 and 3 March 2025 to 15 April 2025, the values for Open, High, and Low prices were recorded as 0. This anomaly occurred because the company's stock was placed under a trading halt and suspension on the Australian Securities Exchange (ASX).

Formulas and calculations applied

Calculation	Formula	Purpose
Moving Average (7-day)	$MA7 = \frac{Close_{t-6} + \ldots + Close_t}{7}$	Identifies short-term trends.
Moving Average (30-day)	$MA30 = \frac{Close_{t-29} + \ldots + Close_t}{30}$	Identifies medium-term trends and reduces noise in price movement.
Portfolio Value	$Portfolio\ Value = Shares \times Close\ Price$	Calculates the total value of an investor's holding at a given date.
Profit % (per trade)	$\frac{Sell\ Price - Buy\ Price}{Buy\ Price} \times 100$	Measures the return on a single trade.
Yearly Portfolio Return	$\frac{Portfolio_{end} - Portfolio_{start}}{Portfolio_{start}} \times 100$	Evaluates annual performance under a buy-and-hold strategy.
Market Capitalisation	$Market\ Cap = Close\ Price \times Shares\ on\ Issue$	Determines the company's overall market valuation.

Price-to-Earnings (P/E)	$P/E = \frac{Stock\ Price}{Earnings\ Per\ Share} \times 100$	Measures valuation relative to company earnings; compared with ASX Avg.

Dataset's characteristics and data types

Column	Data Type	Characteristics
ASX Code	Nominal	Identifies the stock symbol.
Company Name	Nominal	Full legal name of the company.
Date	Ordinal	Trading date in DD/MM/YYYY format.
Open	Ratio	Opening price on the trading day.
High	Ratio	The highest price on that day.
Low	Ratio	The lowest price on that day.
Close	Ratio	Closing price of the day.
Volume	Ratio	Total shares traded that day.
Market Capitalisation ($m)	Ratio	Market value of the company's outstanding shares.
Shares on Issue	Ratio	Total number of shares available in the market.
Dividend	Ratio	Dividend per share paid to investors. For SGR, set to 0, indicating no dividend.
Price to Earnings	Ratio	Calculated as Stock Price ÷ Earnings per Share (EPS). Used to measure company valuation.
Price to Earnings (ASX Average)	Ratio	Benchmark P/E ratio (set at 18.0) representing the ASX market average for comparison.

Charts produced in Excel

Close Price vs Volume

In Figure 1, the combo chart illustrates the movement of Close Price against Trading Volume for SGR between July 2024 and August 2025, with trend lines and annotations highlighting significant price movements, anomalies, and trading halt periods.

The chart shows a strong downward trend in SGR's stock price, falling from the highest price $0.58 on 31 July 2024 to the lowest point $0.10 on 1 May 2025. This heavy drop was due to several high trading volumes, such as 313 million shares on 27 September 2024 and 189 million shares on 28 February 2025, which indicate a heavy stock sell-off. Outlier events are also evident during two trading halts, one from late August to 27 September 2024 and another from early March to 16 April 2025. Following these trading halts, minor recoveries were observed, with the price rebounding to $0.15 on 18 June 2025, supported by increases in trading activity.

These findings show that even though there were some short periods where the price went up with high trading volumes, the overall trend of SGR's stock was still going down. This means that the market did not have much confidence in the stock, and it remained risky for investors.

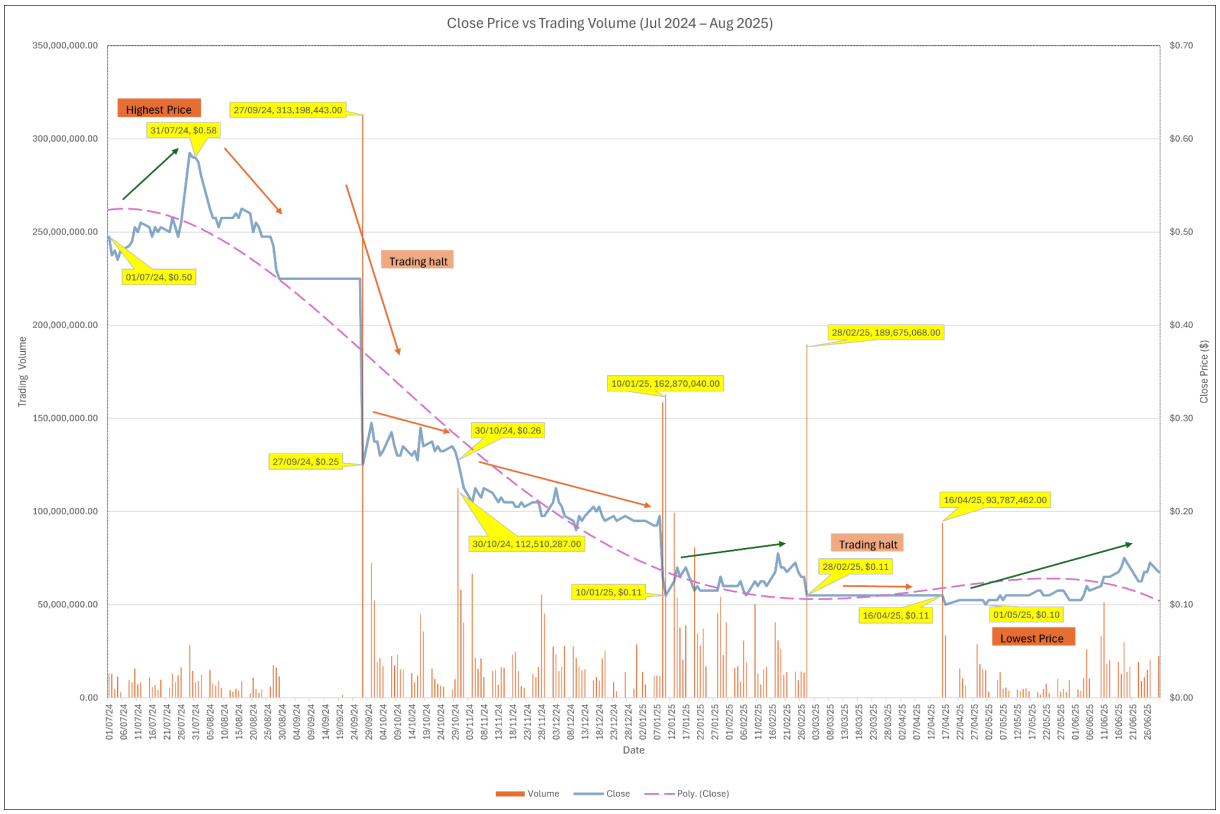

Figure 1: Close Price vs Trading Volume of SGR with Trendline and Trading Halts

Stock Chart

In Figure 2, the stock chart (candlestick chart) illustrates the detailed movement of SGR's stock price from July 2024 to June 2025, with a trendline included to highlight the overall direction.

The chart shows that the price peaked at $0.59 on 29 July 2024 before beginning a continuous decline. A major drop occurred after the trading halt between late August and 27 September 2024, with the price falling to $0.25 on 27 September 2024. The downward trend continued through early 2025, with the lowest point recorded at $0.10 on 10 January 2025. Another trading halt took place from March to April 2025, during which the price stabilized at very low levels. By June 2025, the stock showed only minor recovery and remained around $0.15.

This chart shows that SGR's stock keeps going down and has not really recovered. Even though there were a few times when the price went up, it quickly dropped again. This makes it clear that investors didn't have much confidence in the stock, and the overall trend stayed weak.

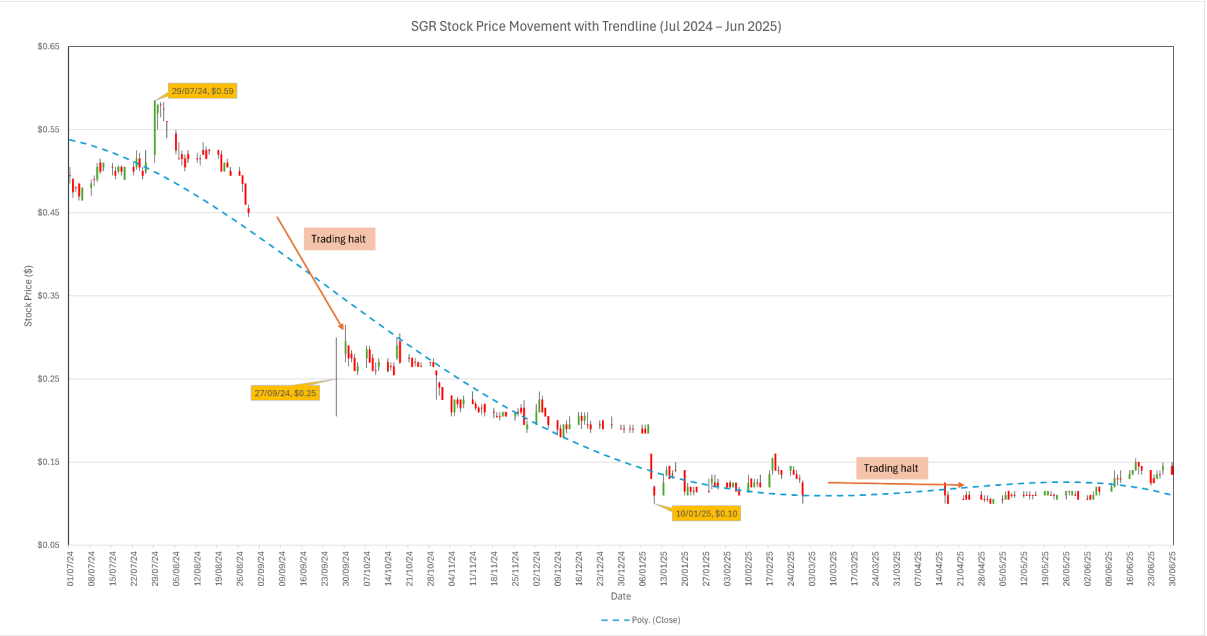

Figure 2: Stock Price Movement with Trendline and Trading Halts

Market Capitalisation vs Shares on Issues

In Figure 3, the line chart compares Market Capitalisation and Shares on Issue for SGR between July 2024 and June 2025. The Market Cap dropped sharply by about 55%, from $1.6 billion to nearly $700 million after September 2024. This decline clearly followed the fall in stock price, while the number of Shares on Issue stayed flat at around 2.8 billion. This shows that the main driver of Market Cap changes was the stock price, not the number of shares. After early 2025, the Market Cap remained low and moved sideways, which reflected the weak confidence of investors.

Figure 3: Market Capitalisation vs Shares on Issue of SGR

Dividend-Yield (Close Price analysis)

In Figure 4, the line chart shows SGR's Close Price trend between 1 July 2024 and 30 June 2025, representing for dividend yield since there is no dividends were paid to the investors.

The stock price started at $0.495 on 1 July 2024, then it reached the highest point at $0.585 on 29 July 2024, but then dropped continuously because of the trading halts and loss of confidence from investors, hitting the lowest point of $0.10 on 17 April 2025. By the end of the period on 30 June 2025, the price had slightly recovered to $0.135.

Overall, the stock price fell by around 72.7% over the year. The 7-day moving average and 30-day moving average also followed the same downward trend. This indicates that since SGR did not pay any dividends, investors can only benefit from changes in the stock price. However, with the steep fall in stock prices throughout the year, investors faced losses instead of gains.

Yearly Portfolio

From Figure 4, the analysis of SGR's stock performance between 1 July 2024 and 30 June 2025 shows a significant decline in the portfolio value. An initial investment of $1,000 at the beginning of the period, when the stock price was $0.495, dropped to only $272.73 by the end of June 2025, when the price fell to $0.135. This represents a yearly portfolio return of –72.7%, meaning that if an investor had held the stock throughout the year, they would have lost more than two-thirds of their initial capital and been left with less than one-third of their starting amount. This clearly shows the high risk of holding SGR in the long term during this period.

Figure 4: SGR Close Price Trend as Dividend Yield Proxy (No Dividend Paid)

Price to Earnings

Figure 5, an area chart shows the Price-to-Earnings (P/E) ratio of SGR with the average P/E ratio of the ASX between July 2024 and June 2025. The P/E ratio of SGR is 22.34, which is higher than the ASX average of 18. This shows that SGR's stock was priced more expensively compared to the overall market. A higher P/E ratio normally suggests that investors expect higher future growth, but in this case, the strong decline in SGR's stock price indicates that the market expectations were not met. This gap highlights that while SGR appeared to be overvalued compared to the ASX average, the company's actual performance did not match the higher P/E level.

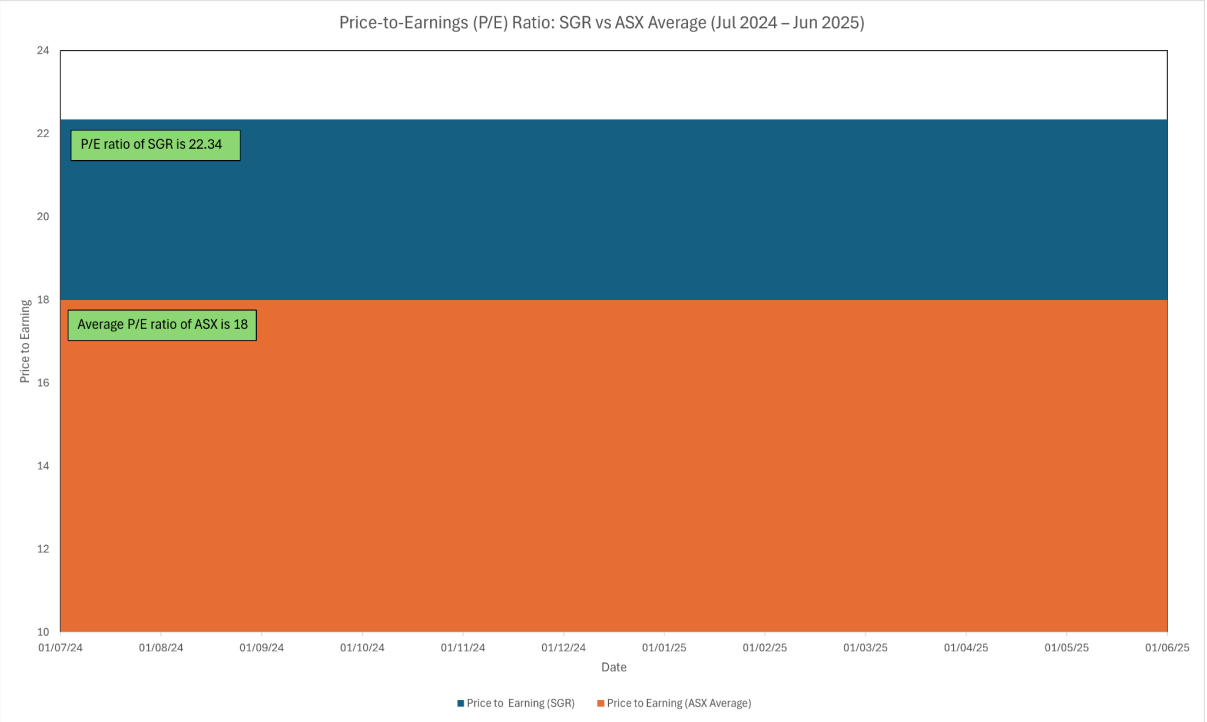

Figure 5: Price-to-Earnings (P/E) Ratio: SGR vs ASX Market Average

Single trade for maximum profits

In Figures 6 and 7, the analysis demonstrates the result of a single trade strategy designed to maximise profit for SGR between the timeframe of 1 July 2024 to 30 June 2025.

In Figure 6, the stock chart provides a time-series visualisation of SGR's stock price, showing how this trade opportunity was identified. The chart highlights the trading halt periods that impacted market movements, followed by the significant rebound. The annotations indicate the buy point at the lowest price and the sell point at the highest price, clearly showing the 150% increase in value during this short period. The trend line in the chart indicates that this point was a potential reversal zone after a long downtrend. The stock price stabilised following the trading halt period, and the rebound confirmed a short-term recovery. This suggests that buying at the lowest point was a logical investment decision, as the trend line supported the possibility of price recovery.

Figure 7 shows the investment value comparison, where an initial $1,000 investment at the lowest price point of $0.10 on 17 April 2025 could have grown to $1,500 by selling at the highest price point of $0.15 on 18 June 2025. This reflects a 150% profit from a single trade.

This figure demonstrates how investors who carefully observe price movements and trend lines can identify profitable entry and exit points even in a highly volatile stock. However, it also underlines the risk that such reversals are difficult to predict consistently.

Figure 6: SGR Single Trade Maximum Profit Opportunity with Trading Halts

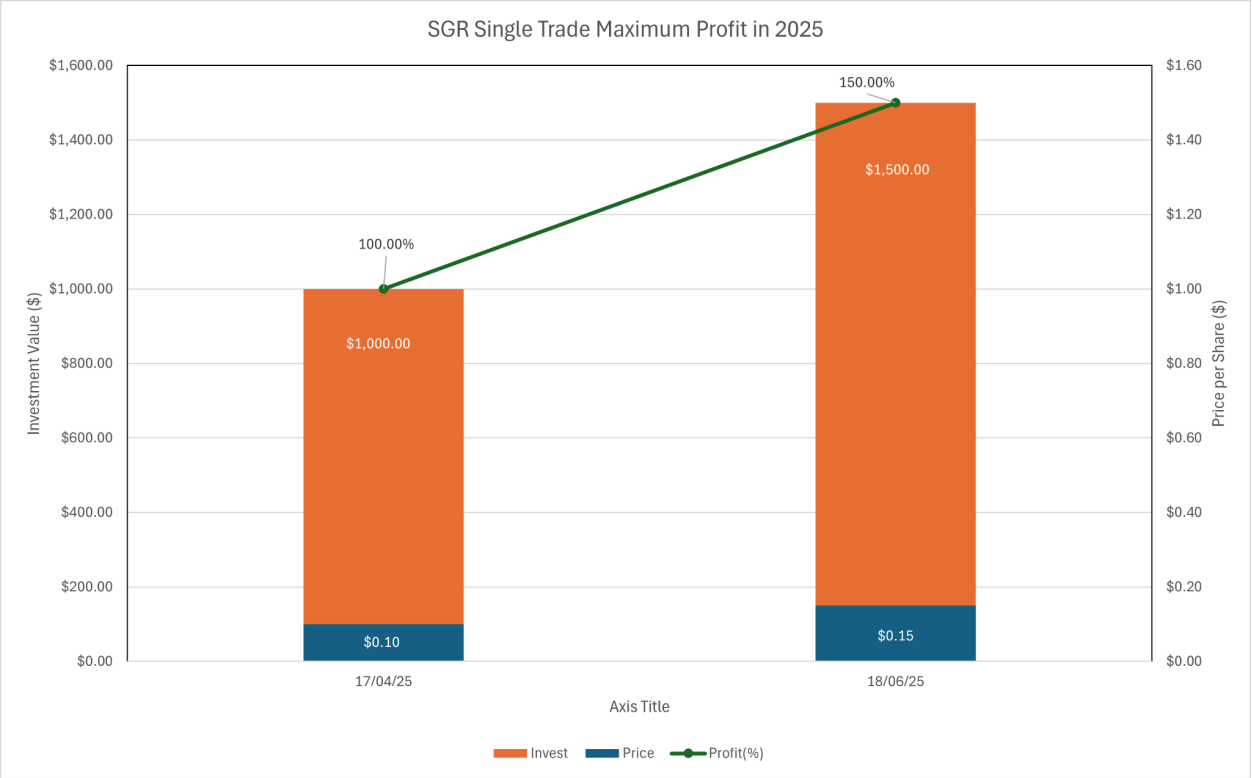

Figure 7: SGR Single Trade Maximum Profit

Multiple trades for maximum profits

In Figures 8 and 9, the analysis presents a multiple trade strategy applied to SGR between July 2024 and June 2025, showing how investors could capture profit opportunities at different points despite the overall declining trend.

The first trade started with a buy at $0.47 on 4 July 2024 and a sell at $0.59 on 29 July 2024, which gave about a 124% return. The second trade was buying at $0.25 on 27 September 2024 and selling at $0.30 on 30 September 2024 for a 118% profit. The third trade happened when the price dropped to $0.11 on 10 January 2025 and later rebounded to $0.16 on 18 February 2025, giving about 141%. The last trade was between $0.10 on 17 April 2025 and $0.15 on 18 June 2025, which gained 150%.

A key technique applied here was the 7-day moving average line. It helped smooth out short-term price changes and made it easier to see when the price was starting to stabilise. This indicates buy signals, which increased the chance of catching rebounds.

Figure 9 shows what would have happened if all four trades were done one after another, starting with $1,000. After the first trade, the investment grew to $1,244.68, then to $1,468.72, then $2,069.56, and finally $3,104.35 after the fourth trade. This means the portfolio more than tripled in just one year.

Overall, even though SGR's price trend was mostly bearish, using simple tools like moving averages and watching for support points made it possible to find short-term profits.

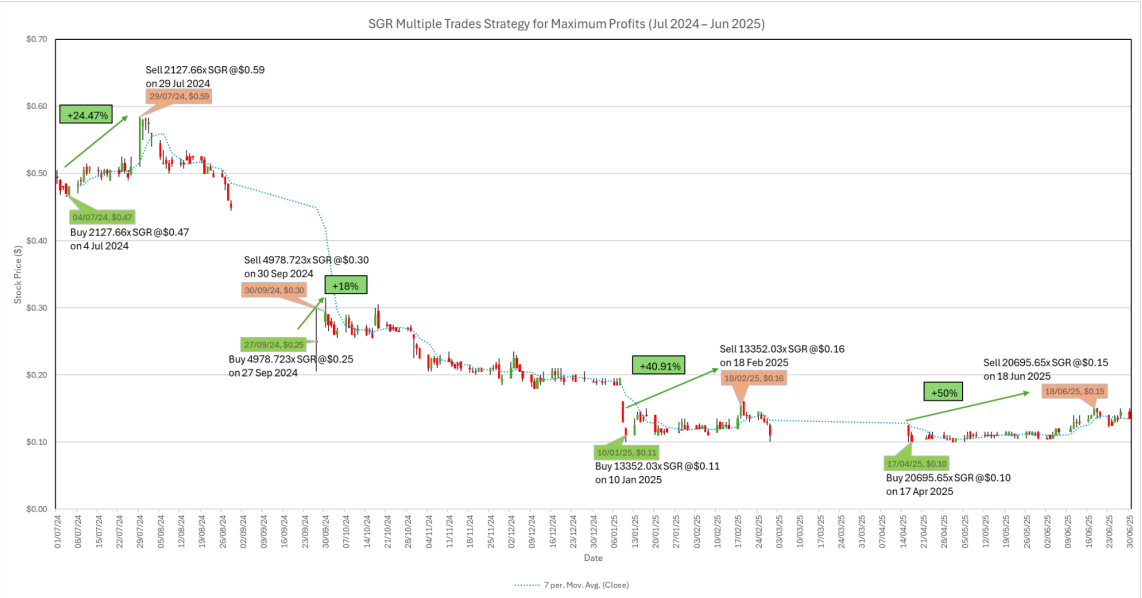

Figure 8: SGR Multiple Trades Strategy for Maximum Profits with 7-Day Moving Average

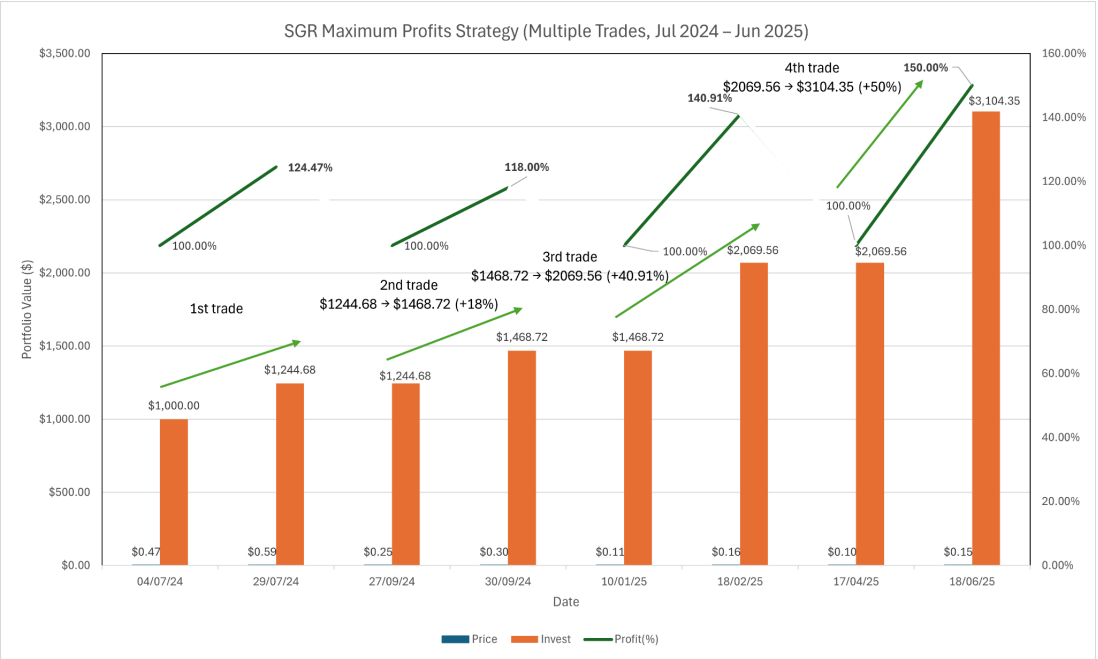

Figure 9: Portfolio Growth from Multiple Trades Strategy

Types of visualisation used

In this report, different types of visualisations were applied to show SGR's stock performance from multiple perspectives.

- Combo charts were used to compare trading volume and closing price, making it clear how heavy trading days are linked to big price movements.
- Candlestick charts with trendlines were used to highlight daily price changes and long-term trends, and these were supported with annotations marking key buy and sell points.
- Line charts with moving averages (7-day and 30-day) were also included to smooth out short-term fluctuations and identify trend reversals.
- Bar charts were used to compare market capitalisation and shares on issue, showing the direct link between stock price and company value.
- Portfolio growth charts were applied to visualise single-trade and multiple-trade strategies, making profit opportunities easy to compare.

To improve readability, I added clear labels for important dates, highest and lowest prices, and profit percentages. Colour coding was used consistently: green for profits, orange for investment values, and red for losses. Arrows and callout boxes were included to guide the reader's attention to significant points like trading halts or trend reversals. This combination of chart types, labelling, and colour design helped create a storytelling approach, allowing the charts to communicate not just numbers but also the meaning behind market movements.

Challenges

When creating the visualisations, I faced some challenges. The first was data overlap, especially with labels and annotations. I fixed this by moving labels outside the data points and using arrows. Another issue was data crossing when different lines overlapped. I solved this by changing line styles and colours. I also had to deal with axis scale problems, for example, when combining price and trading volume. I used a secondary axis to make the chart easier to read. These adjustments helped make the graphs clearer and more focused.

Conclusion

The summary analysis of SGR from 1 July 2024 to 30 June 2025 shows a consistent downward trend in price, with the stock falling from $0.58 in July 2024 to $0.13 in June 2025. Market capitalisation fell sharply even though the number of shares on issue did not change, which means the drop in value came purely from the falling share price. Since the company did not pay any dividends, a buy-and-hold investor would have lost most of their capital, with the yearly portfolio return showing a loss of around 72.7%.

At the same time, the visualisations highlighted short-term trading opportunities. The single-trade strategy showed that a $1,000 investment could grow to $1,500 (+150%) by timing the rebound correctly. The multiple-trade strategy, using 7-day moving averages and trendline analysis to find entry points, showed that compounding profits from four trades could grow to $3,104 (+310%). These findings suggest that while SGR is unsuitable for long-term investors, it may still appeal to short-term traders with a high risk tolerance who can monitor technical signals closely.

The use of different visualisation types was crucial in telling this story. Combo charts made it clear how high trading volume linked with sell-offs, candlestick charts highlighted daily movements, moving averages identified turning points, and portfolio growth charts showed the real financial impact of different strategies. Labelling, colour, and annotations were applied to guide the reader’s attention to important points such as trading halts, lows, and rebounds.

One limitation of using AI alone is that it often gives results without explaining clearly how those results were made. Investors may get predictions or signals, but they cannot always see the evidence behind them. This can make it harder to build trust in the analysis.

By creating visualisations manually, the process is more transparent. Every trend, trading halt, or rebound can be shown on a chart, with labels and annotations that explain why it happened. This makes the information easier to understand and allows investors to connect the numbers with real events in the market. In other words, AI may be faster, but visualisation provides the story behind the data, which is just as important for making investment decisions.

In conclusion, SGR shows weak fundamentals and high risk for long-term investors, but short-term trades supported by technical indicators may still create opportunities.